

Search sheets: how to run the activity



The pages after this one are 36 **search sheets**, built from *The Cat in the Hat* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

If the room is a different size, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 54–55 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **in** **the**, the boxed token is the previous token and **the** is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 8 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *the*? it’s a purple one” — and most of the page stops being worth reading. The swatches are chosen to answer to these words, so the words are the ones to use; the sheets themselves print no key, because a strip of swatches at the top of a page of swatches is one more thing to scan past.

black **grey** **red** **brown** **green** **blue** **purple** **magenta**

Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Quiet rounds. One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

1962 tokens, 240 unique · 2.83 bits/token · perplexity 7.1 · branching factor 4.32

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token *and* its next token on the board.

Round 1. Call out the last token on the board (*the*). Another sheet has a match, so that person raises a hand and reads out *milk*.

Round 2. Call out the last token on the board (*milk*). Another sheet has a match, so that person raises a hand and reads out *on*.

Keep going for as long as you like. The text the room writes will sound like *The Cat in the Hat* without ever repeating it — that is the model generating, not the corpus being read out.

the token pair that answers

in the

the milk

milk on

on the board

in the

in the milk

in the milk on

in the game that fish in said ,
 things will Why do what would . that
 kites on tell me like , did not
 the fish is not can have so ,
 a pot fly kites find this I will
 the ball tip of sad kind you some
 fan ! us this , wet ! so
 have to do as should not ! look
 wet to had to saw them with the
 and I bump on . have I do
 he said we do that is milk on
 the way . and then I and so
 to you ! then not mind , I
 will show on a . not

find us did you Why , not all
 ! and and Thing day . and the
 so tame , white said the hands with
 a cake , what ! said is a
 shame ! wet day kites , I call
 . . . your with my then we
 can hold put them I do . your
 we had have Lots . he tip of
 to us the milk will not saw him
 you do the cup mess , on the
 fish , like to he put fan as
 do something house and ! said
 that is the fish milk and him !
 what to call up about it things and

then , him step mat ! fan ,
put me . you what will house .
? Oh you can call them on a
I said deep and wet , the cat
! and can hold to us . we
and Sally have any ! Thump hop up
with a things I ! Who , I
said the saw him is not the cat
the books mess is , I toy ship
and I I know we saw . and
he said like it hands . milk ,
not a will see . he about with
that are day . shame ! fish !
so so bad tricks do !

what to	him pick	! said	is not
the hat	. Yes	fish .	. two
house !	call this	with a	, with
are pink	said the	want to	toy man
so all	put them	shook their	hands with
you ?	will like	! look	fan with
bad ,	day .	wet and	like these
can hold	? Well	shame !	mess !
I will	! so	on there	he fell
we did	and so	do ?	about .
not know	that is	then Sally	. that
the hat	saw one	to find	and big
things fall	. do	hop on	, said
I will	the cake	deep in	

! what	things they	our fish	get them
gown with	fish in	then Sally	. you
wet day	house !	lit .	! and
the toy	toy ship	not sunny	to us
want to	day ?	I can	some good
we looked	he gave	on his	that bump
would you	what a	and Sally	. so
put me	so .	do not	I say
! now	good fun	will show	saw him
the cat	Think of	, said	a ball
can hold	said to	here .	like it
with a	the house	shook up	, no
you should	? Why	and we	are two
. .	there like	. Thing	

. then our fish ! and good ,
 on a house ? and down . I
 Sally . shook with Lots of get out
 what the the hat a cup can hold
 things yet some fun are good there with
 fish as not all the cat he said
 so we ? tell toy ship ! I
 . But will not . our From up
 and Thing to shake would she , and
 I do , we put them I wish
 you hear like to said the yet !
 then I gown ! that . we saw
 the hall kind of now you ! bump
 here When not be at me

Sally and pick up that is ! and
 , how here When so big you sank
 their two things in like it hops and
 are tame good games not like .
 what did , and said the get those
 then those fish said ! sit they will
 house ! our toy not all Lots of
 to be the cat we saw .
 ? and a cup he came some new
 . I on the and Thing ! he
 . and I like there , would you
 the cat at all gown , now look
 bit ! of her can hold I said
 and this will she the hat

has gone then , here is things that
and the he was I will a bow
could see there that we had so .
will hit be in at this not like
I will . what . my ! then
can get that is ! we they will
their kites bit ! on the our mother
said the not be ! you some milk
are not fish said pick up and the
? I like that good tricks what to
now ! you wish . I , wet
the fish of the get rid the cat
house ! to get , too bite you
. we Sally and the cake

tail I that is good trick he lit
gave them things . like to dish !
of this be here cold to can fan
! then and the PLOP ! now !
what you on my not . , not
here to they play we saw . .
? what at all ! said then our
not shine house ! a pat hear !
our mother to fall there on you up
could , the rake the cat said the
fish . pick it I can I will
. he so we . But , said
. and Sally and will have For the
bit . ! those the fan

out he got to do into a
hear ? ! the a look , then
! and and I our fish fish shook
! what were down pick up dish ,
things . I will not in could do
Sally and there is the hat be here
here . at me on the the fish
they like ? should in the will show
how do fast as now ! house all
! she that were of this .
cold , I do not fly the hook
said , pack up can get ! Oh
like this what will good game .
we can then he you pack

we saw a rake fast as of good
in the can hop will not . he
be here kicks and not be books !
up and ! I on the man !
and all , she things ran I can
could play our fish kinds of give you
, what that I cold , . we
fish . you sit ! I cat go
. and string of house For how that
what would I hop the cat the dots
? Oh good things said to to know
then something now , here When
he shut . . at that Sally and
like the the house ! look they like

said to

can have

. these

be here

, we

things they

? now

one and

in with

two things

the day

what a

and the

not want

how I

on the

at me

will give

strings ,

to the

a book

then he

! so

and Thing

! if

box ,

I ,

now look

playthings and

books ,

like it

up these

of something

the pot

mother's new

! they

. then

. and

our house

fish in

. I

we saw

net .

wood box

fast .

, so

you like

that I

lot of

I can

show you

cat in

the ball

they are

good fun

I call show to I can box are
like our one little the house our new
a tip , said ! said to say
you another trick that little toy , the
! her now , net . in the
! put out of , Why . this
cat in lot ! fish said what a
of good . it . and they should
pat . two books . Oh good game
things that said the not be we sat
how . will not the box play .
books ! at all be here and kicks
then he fast as that day and so
? would up the the fish

, with . so . I to the

the things then he trick , fast !

out and we can net , last !

will come looked ! they should the hat

Thing one and a Oh , two things

! do I can fish came pot !

tricks . one and the fish cat in

said the . a in the , do

a box our mother and look now .

at last ! and be about box .

I do you have ! said that is

up on , said show them not one

came down play ! no fear little fish

. this ? Oh of look

no , . I I said show you
up those box with you do we sat
. have pot . will bump the cake
then our the mat two , Oh ,
that they out ! little bit not wish
Thing two . the ship and red
looked ! now look one one ! at
, no said the of his in the
, and and I net I funny !
sat in , here they will cat .
came in our fish at me ! those
hall . ! put sun did a bump
fish . play , tricks . to go
. they I can ? and

of Thing Oh dear and the . there
our fish box . in the . I
little bit ! how ! then two .
them go ! then not . one and
. with cat in said , play ball
pot . , what tricks , no fear
Thing two bow . and you hall .
out ! take them will find the cat
we sat this mess now ! came down
sun is you get they want , the
up up look at fish on I can
if your book one at this the wall
the sun . that I let a lot
, I to do sat there

, jumps not pick . you our game
look ! if you to do and the
in a and a take a fall .
When your of one , how do not
we looked they ran play . ! said
the hook the cat little bit two things
fear . . I the milk at all
this mess said the came down , look
a big them a no fun I do
out of ! have home ! I saw
box came ! I Oh , one hand
up high pot . run down sat there
Thing one his hat cat . tricks are
you will fish to . it

up . at me ! they cat .
 When our in the the cat came Thing
 , now of the hand ! the cat
 and Thing . . and with Thing
 to the , I sat in no fear
 you now we tell Oh , the wall
 those things I call new tricks play .
 do ? they have look at asked you
 one ! his hat I always said the
 fear ! them out , said . no
 out ! fall ! Thing one if I
 not know ! then little toy a little
 pot . ! said bed ! two things
 and I this game

Thing one new . at this a tip
 came back this way Well . we saw
 the house , and out of not be
 . they When your said , cat .
 went on and Thing his head , said
 and they I had fall ! look at
 no fear ! I fear ! ! now
 . I . in is what ! he
 Make that one ! up all the cat
 with a you do of my do .
 Oh , tame . was gone thumps and
 them out the hat shake hands in the
 I can fell on they ran those things
 if mother two and

cat with said the out . a fish
! put . . said our tame !
those things ! and I bet went away
cake ! I know Make them Oh ,
another good it was they should
new rake up all fear , shake hands
Thing one know how no ! , that
the fish and the fell into . I
wall ! with big not know do not
 . you look at this house and you
the things ! and of his in the
 , said them I one little two !
is not the fish was out mind at
you did say . we could

! they , said sank it Oh ,
 my tricks cat , Thing two .
 the things was to say . and our
 they said two ? the hat . and
 the pot , and up the a box
 with hops is not fear of sit !
 said the of good I stand went bump
 look ! bet . ! have ! sit
 I did new gown another good
 one little and the it , this trick
 those two know what mother was not one
 you . all the down on do not
 them Thing wall in in the cake .
 said the . I no fear

one kite no ! do . as I
 Oh , they ran all . me !
 I sat my tail . they look what
 them to the cat sank our not let
 . should fear . said the ! bump
 the cat come here , as and he
 is not , sank ! it went after
 was too her the two things and so
 up there wish we with a say .
 sit ! . your cake , know !
 I like cat . of bad those two
 it deep bet , in the Thing two
 said , this wet ! not down the
 mother could a red

was shut . so then come .
those things ! what sit ! say ?
step in ! and is good up our
cat went wish to and some I know
the string down with me . them away
bumps , one kite I went , no
a dish two . do you as I
cake on mother is look ! with a
to have , and ! your the hat
big red not ! hat . this box
all kinds fun to and our . now
it a my playthings no !
her . said , Thing two know it
Oh , way that in this

the fish	the cat	to play	cat !
and look	. and	not bad	and he
said ,	in the	away .	it was
a sad	all we	! and	them away
sit there	way !	high as	my hat
her way	mother is	, if	, said
! down	Oh ,	you will	do I
is near	fun in	no way	as fast
I could	was back	Who was	gone with
down my	up up	say !	one and
me now	! said	. you	hat !
two .	this is	big and	with a
Thing one	know some	. I	wish ,
pink ,	look at		

picked up

, we

to us

, wet

mother !

is wet

. look

as a

you take

gone away

no !

up !

! look

her about

look at

this !

them run

two and

the house

and the

fun on

a house

I did

. Oh

down !

hat you

Oh ,

away !

cup on

know !

! said

! Make

the cat

ship ,

Thing two

me !

said .

my net

But that

these things

wish to

way home

it came

all !

and he

. with

nothing at

in and

cat .

sit !

not bite

do you

big bumps

with a

mother will ! fast a game all .
, and up with tall , stand on
fun at ! said hat . know what
do not in the way at them !
big thumps said the not like no .
two hands cat in and the the hall
I will dear . . Oh . But
me down her bed the gown after my
is fun my net these two cup and
with a stop . . a it is
Oh dear this ? you hear , no
But you and the away . look !
Yes . down the to play ! But
ship ! as I

look it! I like down!
 do was is that dear! fun?
 no away! and then have
 that cat something went my net
 a good , said . hat.
 ball! her gown ship, said the
 But your look! top of up the
 big mess you shook these things as you
 mother asked to say all!
 have to . not one the top
 cat. me down know some ! we
 let you should we with the cup,
 the cat and I , did back in
 in a rake. this is no!

this house should not a game fun But
 it is , said have no ship and
 my net hat . ! and me !
 no . down . let down cat in
 ! said the fish she will ! I
 know what look at all ! and I
 . and away with ! look you bent
 back in up two and we as I
 these books in the not wish cup and
 hit ! I did But I . then
 said the is not mother will do you
 the cat top with , we rake ,
 with a ball . to stop . I
 that they something to

in the ! and all . ! then
ball . ran up to play a shame
my hat and the hat to , as
! and is so did not rake !
kite bump this , you . she do
hook . look at that is down ,
! we up the fun that and Sally
your mother me now should not sunny .
cat . with the as he no !
said the the things have no I can
! . it to But that do ?
something new ! too go !
! they not all the sun any fun
mother came away . , with

so, the cat go away . now
something to should not do !
! take and she , Oh hold up
the box in on she say hat ,
kite we cat in ball . up the
Thump ! have fun But our hook .
! I ! sit with my I saw
got up it . mother is bump !
ran down a PLOP said the is on
did nothing to fly down with .
! look ! I fun , look .
not . your mother , and that I
you do this house as fast he should
too . all the and a

But little have no and
 bump made it and I is out
 fun . in the to you do not
 all . he picked ! on with a
 should not that ! so deep did you
 I know too wet it ? we can
 they she will cat . ! then
 ! put mother is your mother games we
 you do not like and a hook .
 look at a little something ! ran to
 Thump ! the things always pick head !
 , and go away said the ball !
 down ! But that this big hat !
 the cat hold you

and then , cold not like ! no
things had said the it up you fall
will hold all I jump ! cat in
ran about he should game that the pot
on top near ! to sit with Sally
! Oh hold up should we she say
! you bump ! your mother . Oh
did not go out that is mother is
and with . he . he hat .
we shook shut with . now so so
too cold head of in the , and
then we ! and have Lots do to
us two I said But we a shame
is no the ball

dots that things bump hold a and ,
 will show us jump cat in to have
 then he she said too wet saw all
 game fun on her your mother do !
 said the the cat ! they have no
 a fox ! tell bump From you have
 should we with my , fast . you
 we two red . hat . he should
 , no is no . you they
 I bet . But go ! ! But
 and the the fish in the jumps and
 that ? shut up But our did not
 not like the things so we ran out
 all that ! with

will pick should not your mother us ?
tell her do you I do to shake
bump ! on the with fear you another
then ! the fan in this that cold
game . red wood ! Make , I
fox , fish in and too cat .
have fun not be had something
a hook hold up said the , Think
rid of is not we did ! Oh
made us . that ! said see this
. I he fell so tall ! Thump
saw those the fish things should
can not hat ! . then all if
did ! the cat and then . Why

not go white and the hat that cat
your mother ! and cat in I know
fish . and the then he will she
so do all my he ran ! they
did not on this said , to play
tell her , the the strings bent our
red fan should not ! I can ,
us fast . then . Oh and I
hold up the pot wet to . then
a shame hat came game , bump their
you did had them do if have no
. I saw mother's with a
things are see something we could
is funny in the ! Oh , said

so , have come on the . these
 , so the head and I I can
 . what do not and then hold the
 cat said kites in he has ! But
 bump ! we tell should not us ,
 not all Why , what should saw her
 tell that things , ! not to go
 fly kites ! now game that fish !
 , the did not that went can do
 . no then we said , shine .
 with Thing the dish like it in the
 will not a lot the cup . so
 you will is not your mother had to
 hat . wet ,